Two-player handheld: reference design (REF-A)

RP2354B, 3.5in 320x480 IPS (ILI9488, SPI), 16 membrane contacts under DS Lite silicone,
PWM audio into an FM8002A, one Li-ion cell on a TP4056 charger with load sharing, ME6211 3V3.

Every sheet is generated from board/design.py by tools/build_sch.py. Edit design.py, not the sheets.

USB_Charger

usb_charger.kicad_sch

USB-C sink, ESD, TP4056 charger, load sharing, battery connector

Power

power.kicad_sch

ME6211 3V3 LDO, power button latch, battery and button sensing

MCU

mcu.kicad_sch

RP2354B (2 MB flash inside), 12 MHz crystal, USB series resistors

MCU_Power

mcu_power.kicad_sch

RP2354B decoupling and its internal 1.1 V switcher

Debug

debug.kicad_sch

SWD and UART connectors, BOOTSEL and RESET buttons

Display

display.kicad_sch

50-pin ZIF connector for the 3.5in IPS panel, backlight current sinks

Audio

audio.kicad_sch

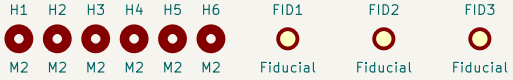
PWM low-pass into an FM8002A bridge amplifier, speaker pads

Buttons

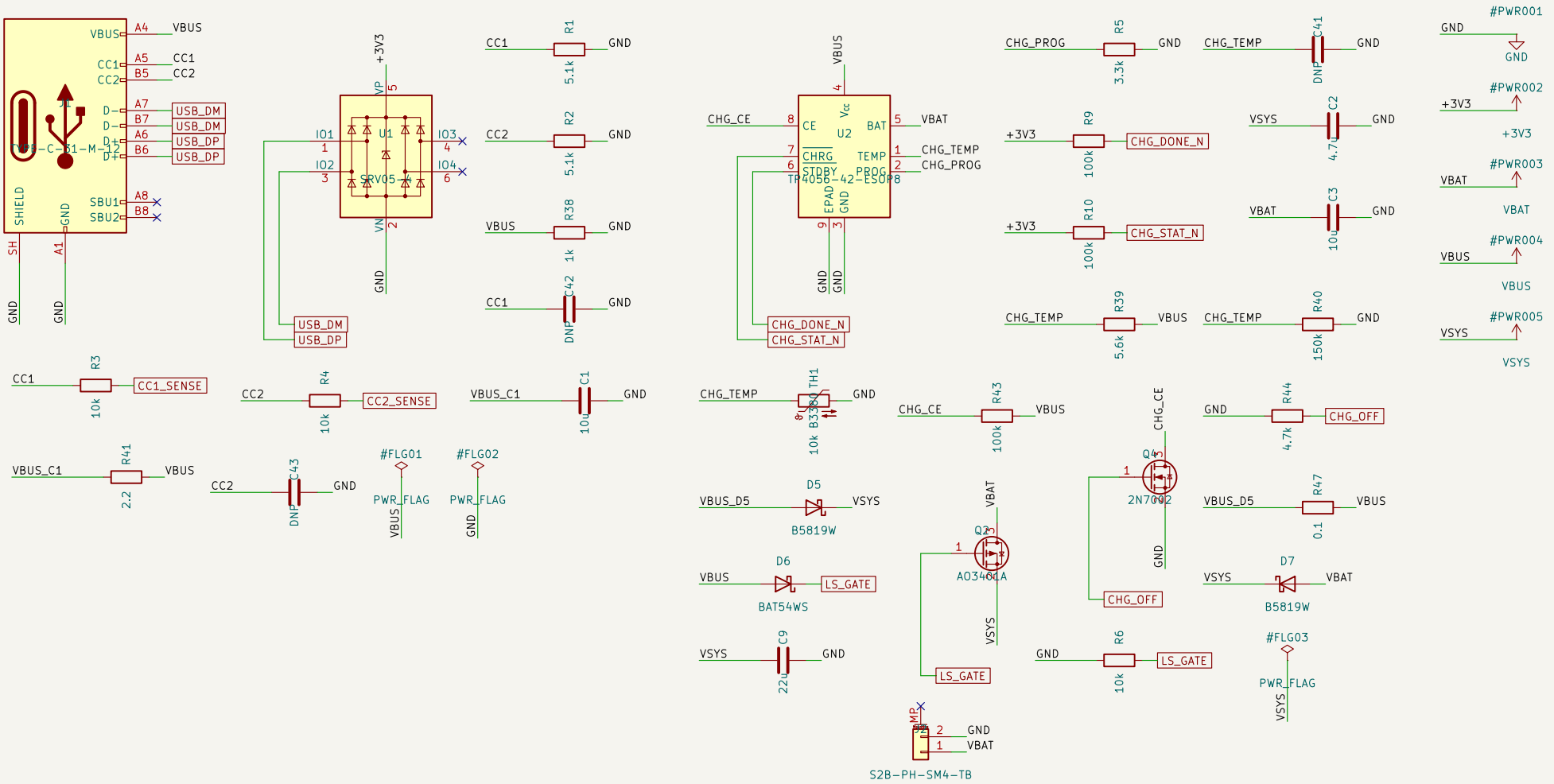
buttons.kicad_sch

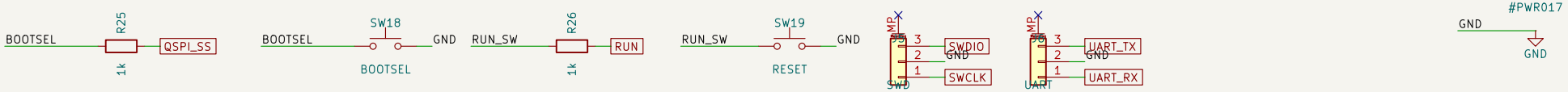
16 membrane contacts, one GPIO each

Mechanical: locating holes at positions set by the case; fiducials for assembly.



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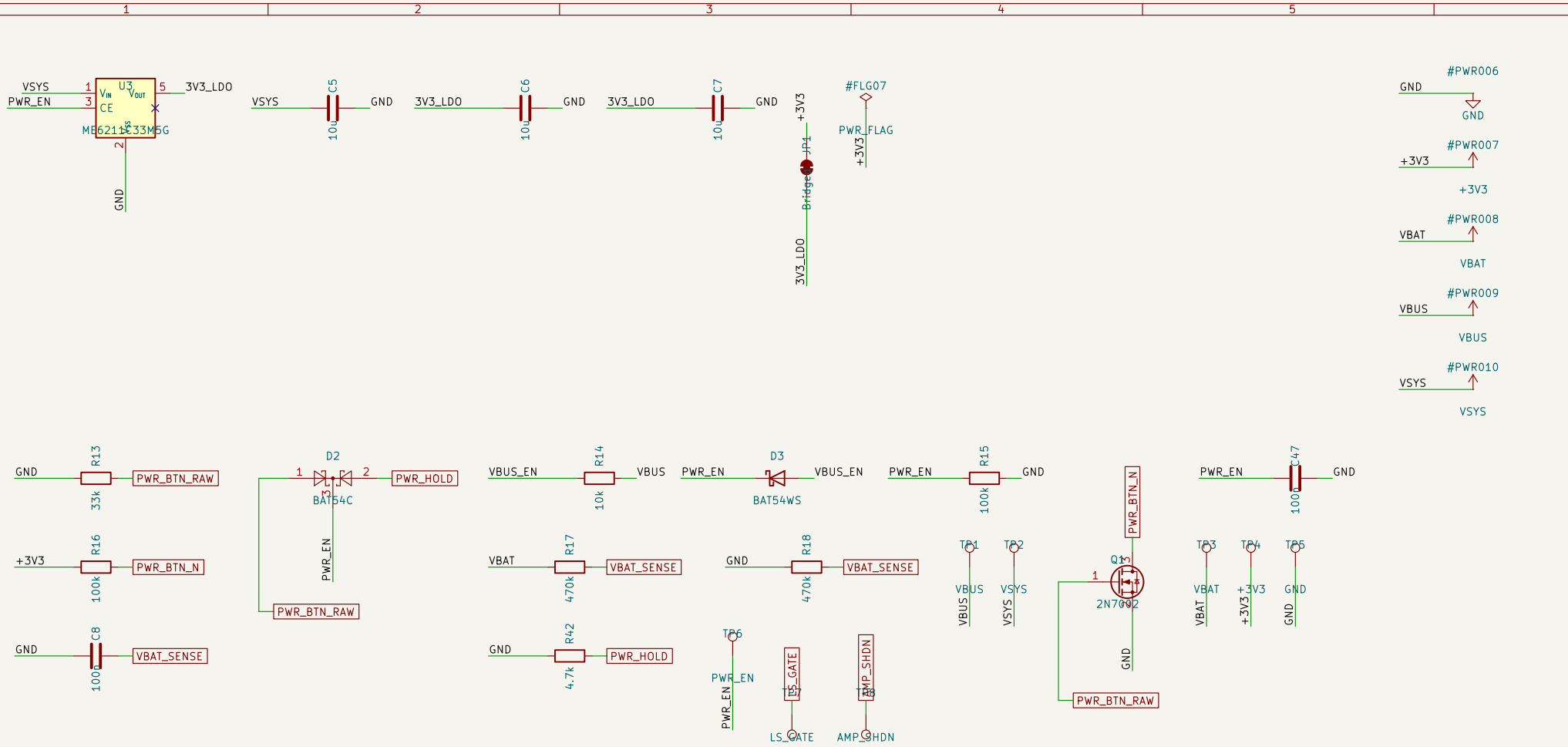

TP4056 charges at 500 mA (R5), from 1 to 46 C (TH1 over the cell). USB feeds the system through D5;
without USB the cell does, through Q2; on an unplug D7 carries VSYS until R38 has drained VBUS and Q2 is on.
CC1/CC2_SENSE let firmware read the source's current offer (dim the backlight on a 500 mA port).
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J5 SWD and J6 UART follow the Raspberry Pi 3-pin debug connector pinout (Debug Probe cable).

SW18 BOOTSEL and SW19 RESET sit under pinholes in the lid.

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Power: the button (seat 1's Y contact), the MCU (PWR_HOLD) or USB raise PWR_EN. Releasing it with PWR_HOLD low turns the device off. A reset drops PWR_HOLD, so a crash turns it off.

Q1 lets the MCU see the button (PWR_BTN_N low = pressed) without exposing a pin to VSYS.

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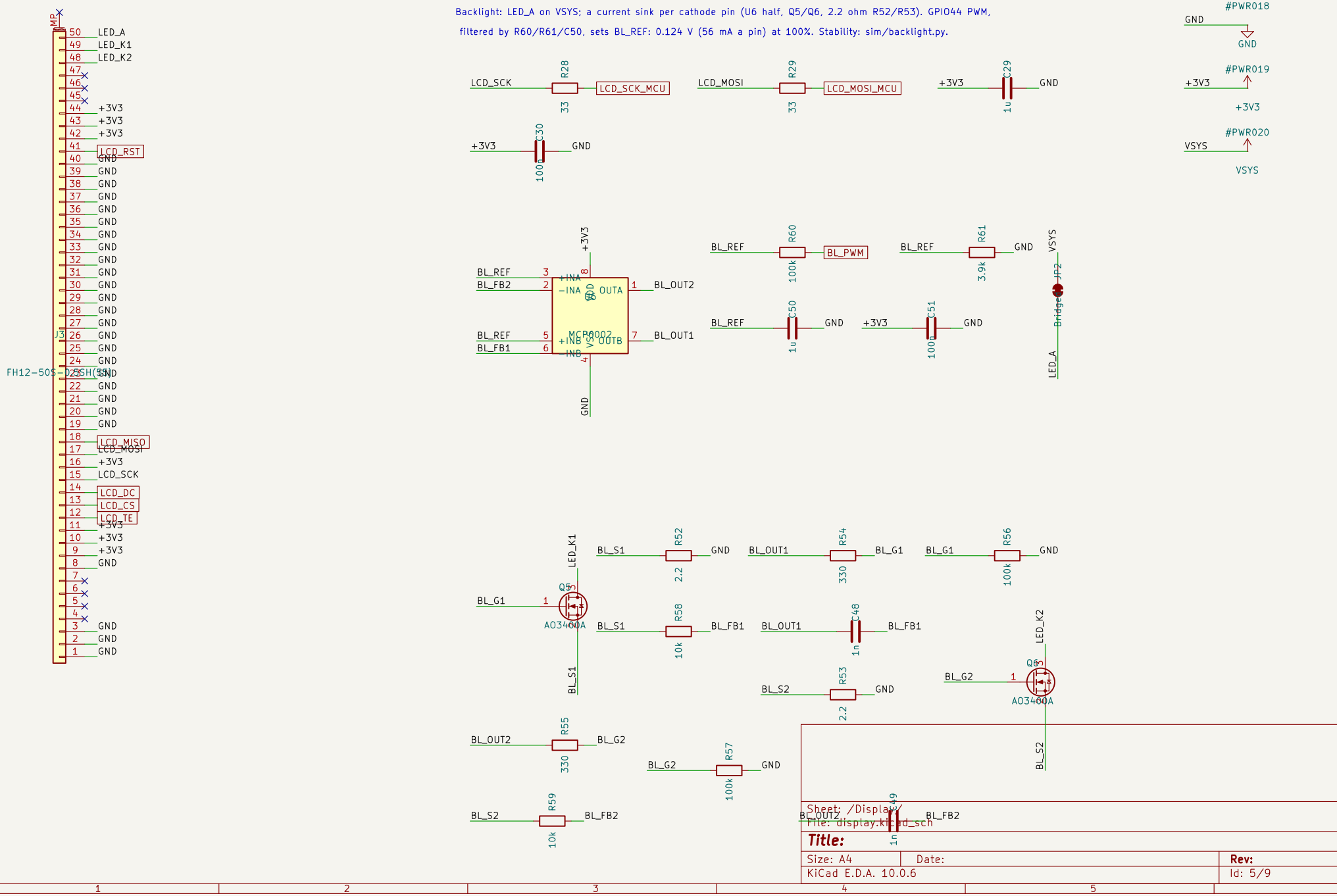
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EastRising ER-TFT035IPS-6 (ILI9488, 4-wire SPI). Its 50-pin FPC folds under the panel into J3 (ZIF).

J3 pin j is panel pin 51 − j. Unused bus pins as the datasheet says: DB0-17 and RGB low, RD and IM2-0 high.

Backlight: LED_A on VSYS; a current sink per cathode pin (U6 half, Q5/Q6, 2.2 ohm R52/R53). GPIO44 PWM, filtered by R60/R61/C50, sets BL_REF: 0.124 V (56 mA a pin) at 100%. Stability: sim/backlight.py.

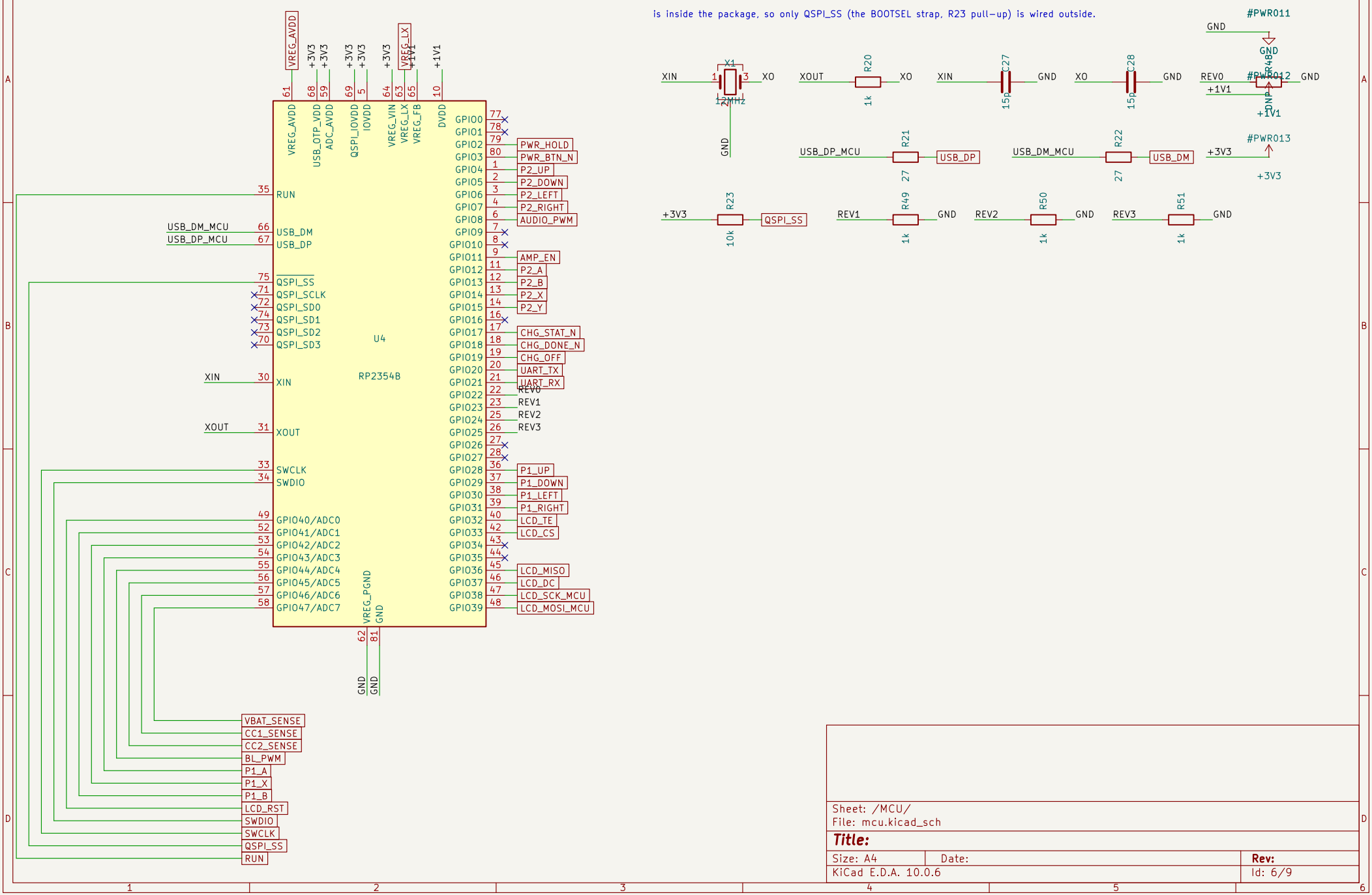


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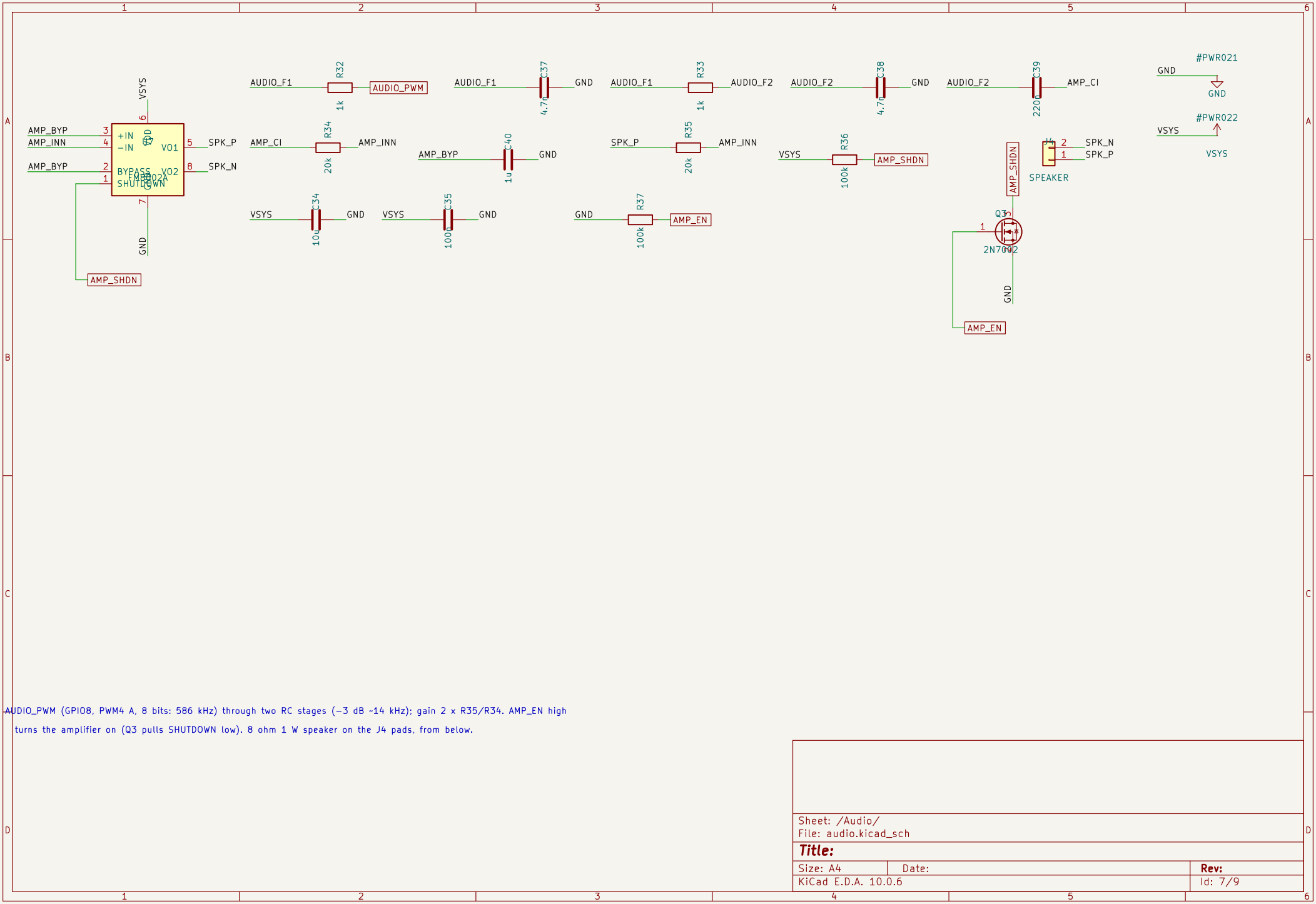
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Core from Raspberry Pi's RP2350B minimal design (MIT), fitted with an RP2354B: its 2 MB flash is inside the package, so only QSPI_SS (the BOOTSEL strap, R23 pull-up) is wired outside.

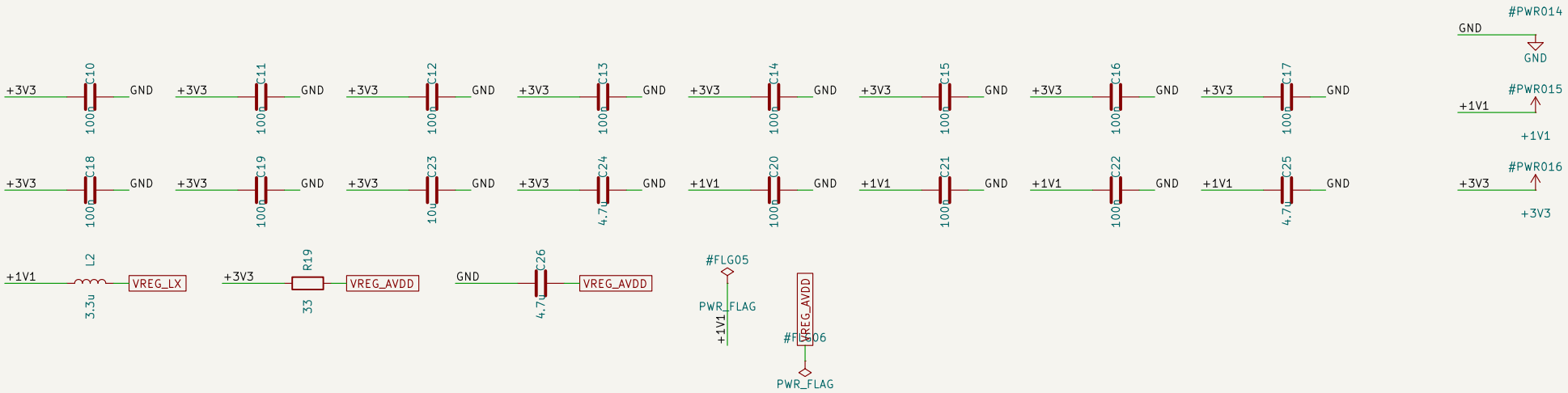


Sheet: /MCU/		
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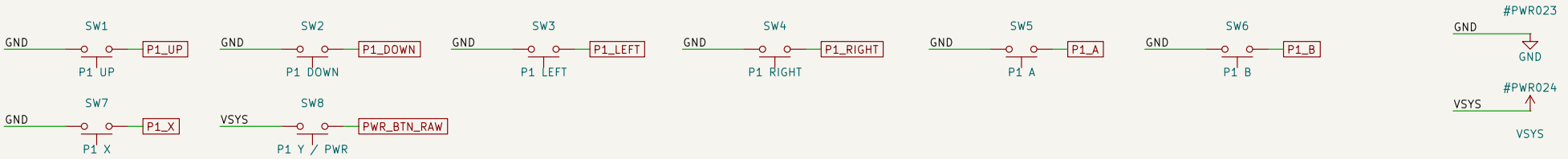
AUDIO_PWM (GPIO8, PWM4 A, 8 bits: 586 kHz) through two RC stages (−3 dB ~14 kHz): gain 2 x R35/R34. AMP_EN high turns the amplifier on (Q3 pulls SHUTDOWN low). 8 ohm 1 W speaker on the J4 pads, from below.

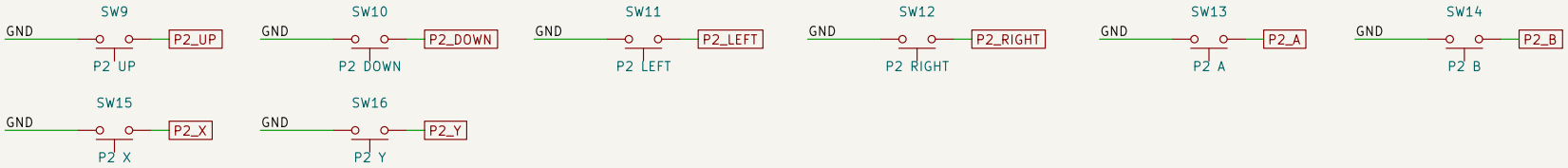
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RP2354B supply, as in Raspberry Pi's minimal design: 100 nF at every IOVDD, DVDD, ADC, OTP and QSPI pin, 4.7 uF at the regulator's input and output, and L2 (a polarity-marked 3.3 uH inductor, fit as drawn).

Sheet: /MCU_Power/	
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Bare ENIG contacts (DSL_button_4ring) under DS Lite silicone; the firmware enables pull-ups.

Sheet: /Buttons/		
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